PROJECT DOCUMENTATION

SMART HEALTHCARE DISEASE ANALYSIS

A Tableau-based analysis of heart disease, diabetes and stroke

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Prepared using the Smart Healthcare dataset supplied through the SkillWallet capstone workspace.

1. INTRODUCTION

1.1 Project Overview

The Smart Healthcare Disease Analysis project uses Tableau to explore patterns associated with three major chronic diseases: heart disease, diabetes and stroke. The work is organised around three practical stakeholder scenarios: clinical risk assessment, lifestyle analysis and population health research. A dashboard-based approach turns a structured health dataset into visual summaries that can be explored more easily than raw rows of data.

The analysis uses the file smart_healthcare_dataset.csv, containing 5,000 records and 16 fields. The available variables cover demographic information, clinical measurements, lifestyle factors, symptoms and disease indicators. Nine visualisations were developed and grouped to address the three scenarios.

1.2 Purpose

The purpose is to demonstrate how data visualisation can support exploratory healthcare analysis. The project aims to compare the occurrence of the selected diseases, inspect relationships among clinical indicators, examine lifestyle and symptom patterns, and identify differences across age groups and BMI categories.

The dashboard is intended as an analytical and educational aid. It is not a diagnostic system, does not predict an individual's medical outcome, and should not replace advice from qualified healthcare professionals.

Project objectives

Objective	Description
Compare disease indicators	Summarise the recorded prevalence indicators for heart disease, diabetes and stroke.
Explore clinical measures	Review blood pressure and glucose patterns across disease-status groups.
Study lifestyle factors	Examine smoking, alcohol and exercise categories in relation to chronic-disease status.
Explore population patterns	Compare disease indicators across age groups and BMI categories.
Review symptoms	Summarise recorded symptom burden and its relationship with disease status.

2. IDEATION PHASE

2.1 Problem Statement

Healthcare datasets can contain many demographic, clinical and behavioural variables. When these are viewed only as rows and columns, it can be difficult for stakeholders to compare disease indicators, recognise patterns and identify questions that deserve further investigation. The project addresses this analysis and communication problem by building a set of interactive Tableau visualisations around three stakeholder scenarios.

Scenario 1 – Dr. Michael Carter, Clinical Risk Assessment: A clinician wants to review how recorded clinical indicators, particularly blood pressure and glucose, vary between people with and without the selected chronic-disease indicators. The key need is a clear clinical overview that supports further investigation of potentially higher-risk groups.

Scenario 2 – Emily Rodriguez, Healthcare Data Analyst: An analyst wants to explore how lifestyle factors such as smoking, alcohol consumption and exercise level are associated with recorded chronic-disease status. The goal is to compare groups and identify patterns for deeper analysis, without assuming that an association proves causation.

Scenario 3 – Dr. Priya Nair, Public Health Researcher: A researcher wants to understand how disease indicators vary across population groups. The analysis focuses on age, BMI and reported symptoms to help describe population-level patterns and highlight groups that may merit additional study.

2.2 Empathy Map Canvas

The empathy map below summarises the likely needs represented by the three project personas. It is a design aid based on the scenario descriptions, not the result of formal interviews.

Persona	Thinks / Needs	Sees / Does	Pain points	Desired gain
Dr. Michael Carter	Needs a quick clinical overview and comparable indicators.	Reviews blood pressure, glucose and disease-status summaries.	Raw data is difficult to scan; averages can hide individual variation.	Clear comparisons that help decide what to examine further.
Emily Rodriguez	Needs to understand lifestyle patterns in the dataset.	Compares smoking, alcohol and exercise categories.	Multiple variables make comparisons time-consuming; association may be misinterpreted.	Consistent charts that reveal group-level patterns.
Dr. Priya Nair	Needs a population-level view of differences between groups.	Segments records by age, BMI and symptoms.	A single overall number can conceal subgroup differences.	Readable summaries for population-health exploration.

2.3 Brainstorming

During ideation, the project requirements were translated into visual questions. Bar charts were selected for category comparisons; a scatter plot was selected to inspect the relationship between two clinical measures; a highlight table was selected for compact demographic summaries; and grouped category charts were selected for age, BMI and symptom analysis. The nine visuals were then organised into three scenario-focused dashboard groups.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Stakeholder activity	Information need	Dashboard support
1. Identify a question	Clinician / analyst / researcher defines a question.	Which disease or population factor should be examined?	Scenario-specific dashboard sections establish the focus.
2. Locate relevant data	User reviews available variables and categories.	Which fields represent disease, clinical, lifestyle or demographic information?	Dataset fields and calculated categories organise the information.
3. Explore patterns	User compares groups and measures.	How do indicators differ between groups?	Nine charts provide prevalence, clinical, lifestyle and population views.
4. Interpret cautiously	User checks what the visual does and does not show.	Is this a descriptive association or evidence of causation?	Labels, aggregation choices and limitations guide interpretation.
5. Communicate findings	User presents the analysis to others.	Which visual answers each scenario question?	Dashboard layout and project explanation video support communication.

3.2 Solution Requirement

Functional requirements

- Import the supplied CSV dataset into Tableau.

- Create calculated fields for chronic-disease status, age grouping, BMI category and symptom burden where required.
- Build nine visualisations covering the three scenarios.
- Arrange the visualisations into a dashboard experience that allows the user to compare relevant groups.
- Save the Tableau workbook and publish a shareable dashboard for demonstration.

Non-functional requirements

- Charts should have readable titles, labels and legends.
- Aggregations should match the meaning of each variable; binary indicators should not be summed when the intention is to compare average recorded status or prevalence.
- The workbook should remain understandable and maintainable.
- Data and conclusions should be presented with appropriate limitations and privacy awareness.

3.3 Data Flow Diagram

Step	Data flow
Input	smart_healthcare_dataset.csv (5,000 records; 16 fields)
Preparation	Connect CSV in Tableau; inspect field types; create derived categories and indicators.
Analysis	Aggregate and compare disease, clinical, lifestyle, age, BMI and symptom variables.
Visualisation	Create nine charts and arrange them into scenario-oriented dashboard sections.
Output	Tableau workbook, published Tableau Public dashboard, screenshots and project report/video.

Text representation of the flow: Dataset → Tableau data connection → Field preparation and calculated fields → Nine visualisations → Scenario dashboards → Published dashboard and project documentation.

3.4 Technology Stack

Technology / Resource	Role in the project
CSV dataset	Structured input data supplied through the SkillWallet capstone workspace.
Tableau	Data connection, calculated fields, chart creation, dashboard layout and publishing.
Tableau Public	Public hosting and sharing of the completed dashboard (publication step).
GitHub	Versioned repository for the workbook, documentation, templates and project links (submission step).
Google Drive	Storage and sharing of the project explanation video and supporting files (submission step).
PDF / Word tools	Preparation and export of the formal project documentation.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The project fits the stated problem because it converts a multi-variable healthcare dataset into a small set of focused visual comparisons. Each group of charts is aligned with a stakeholder scenario: clinical indicators for the clinician, behavioural factors for the analyst, and demographic or symptom patterns for the public-health researcher. This makes the analysis easier to navigate and communicate than an unstructured spreadsheet.

The solution is appropriate for descriptive exploration. The dataset is observational and the charts show patterns in the supplied records; they do not establish causal relationships or validate clinical risk models.

4.2 Proposed Solution

The proposed solution is a Tableau dashboard containing nine visualisations, arranged into three scenario-oriented groups. The first group examines disease prevalence and clinical indicators. The second group compares lifestyle categories with chronic-disease status. The third group explores age, BMI and symptom patterns. Together, these views provide a structured overview of the dataset and allow the audience to connect each visual to a specific analytical question.

4.3 Solution Architecture

Layer	Components	Purpose
Data layer	CSV file; 5,000 rows; 16 fields	Provides the source records for analysis.
Preparation layer	Tableau data source; data types; calculated fields	Creates analysis-ready groups and indicators.
Analysis layer	Measures, dimensions and aggregations	Compares disease status across relevant variables.
Visual layer	Nine charts; scenario-oriented dashboards	Presents patterns in a readable form.
Delivery layer	TWB/TWBX workbook; Tableau Public; PDF report; video; GitHub	Supports review, sharing and academic submission.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The work was divided into sequential phases so that the dataset and scenario requirements could be understood before dashboard construction. The schedule below describes the project workflow; exact calendar dates and durations should be added from the team's actual work log.

Phase	Main activities	Deliverable / status
1. Understand the brief	Review the three scenarios and their questions.	Scenario requirements identified.
2. Access and inspect data	Download and inspect the CSV; identify dimensions and measures.	Dataset identified: 5,000 records and 16 fields.
3. Prepare data	Check field types and create required calculated fields and categories.	Analysis-ready Tableau data source.
4. Build visualisations	Create the nine charts for clinical, lifestyle and population analysis.	Nine visualisations completed.
5. Assemble dashboard	Arrange charts into scenario-oriented dashboard sections and review layout.	Tableau dashboard completed.
6. Validate and document	Check chart logic, labels, layout and limitations; prepare report and demo.	Report and demo materials to finalise.
7. Publish and submit	Publish to Tableau Public; organise GitHub and SkillWallet submissions.	Public links and review submissions to add.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Testing focused on whether the charts represented the intended fields and whether the dashboard could be read and interpreted. The checks below are qualitative validation checks; no benchmark timings, load-test results or measured response-time figures were supplied, so numerical performance claims are intentionally not reported.

Test area	Validation performed / required	Expected outcome
Data connection	Open the workbook and confirm the CSV connection resolves.	Source data is available to the workbook.
Field and aggregation logic	Check dimensions, measures and aggregation choices; avoid summing binary disease flags when comparing average status.	Displayed values match the intended analytical meaning.
Disease comparison	Check heart disease, diabetes and stroke measures in the prevalence view.	All three disease indicators are represented clearly.
Clinical view	Check blood pressure and glucose axes and disease-status grouping.	The scatter plot communicates the intended clinical comparison.
Lifestyle views	Check smoking, alcohol and exercise categories against chronic-disease status.	Category comparisons are legible and use consistent logic.
Population views	Check age groups, BMI categories and symptom groupings.	Population and symptom comparisons are visible.
Dashboard layout	Review titles, legends, spacing, labels and fit on the target display.	Charts remain readable without unintended clipping.
Sharing and portability	Reopen the saved workbook and verify the published link after publishing.	The project can be reviewed by another person.

Data interpretation checks and limitations

The field health_risk_score was observed to have a constant value of 100 in the supplied dataset. Because it does not vary across records, it cannot distinguish between risk levels and should not be interpreted as a useful ranking or predictive score in this project. The analysis therefore relies on the disease indicators and other recorded variables.

The disease and symptom fields are treated as recorded indicators. A visual association between a lifestyle category and disease status does not demonstrate that the lifestyle factor caused the disease. Aggregated charts can also conceal individual-level variation and should be interpreted as descriptive summaries.

7. RESULTS

The completed Tableau work contains nine visualisations grouped around the three scenarios. The table below records the intended role of each visual.

No.	Visualisation	Scenario	Question addressed
1	Disease Prevalence	Clinical Risk Assessment	How do the recorded heart disease, diabetes and stroke indicators compare?
2	Clinical Indicators	Clinical Risk Assessment	How are blood pressure and glucose distributed across disease-status groups?
3	Clinical Summary	Clinical Risk Assessment	How do selected clinical indicators vary across age groups and gender?
4	Smoking Analysis	Healthcare Data Analyst	How does recorded chronic-disease status vary across smoking categories?
5	Alcohol Analysis	Healthcare Data Analyst	How does recorded chronic-disease status vary across alcohol categories?
6	Exercise Analysis	Healthcare Data Analyst	How does recorded chronic-disease status vary across exercise levels?
7	Age Group Analysis	Public Health Researcher	How do disease indicators differ across age groups?
8	BMI Analysis	Public Health Researcher	How do disease indicators differ across BMI categories?
9	Symptom Analysis	Public Health Researcher	How does recorded symptom burden vary across disease-status groups?

8. ADVANTAGES & DISADVANTAGES

Advantages

- Converts a multi-column dataset into visual summaries that are easier to compare.
- Covers three complementary perspectives: clinical, lifestyle and population health.
- Uses nine visualisations to address a defined set of analytical questions.
- Tableau supports clear chart formatting and dashboard-based communication.
- The workbook and published dashboard can support demonstration and review.

Disadvantages and limitations

- The analysis is limited to the variables and records available in the supplied dataset.
- The dataset does not provide a time field for analysing trends over time.
- The constant health_risk_score value of 100 is not informative for ranking or prediction.
- Descriptive associations cannot establish causation or provide a clinical diagnosis.
- Aggregated summaries may hide variation within categories.
- The dashboard's conclusions depend on the quality, definitions and representativeness of the source data.

9. CONCLUSION

The Smart Healthcare Disease Analysis project demonstrates the use of Tableau to explore a structured healthcare dataset through nine visualisations. The charts are organised around three stakeholder scenarios: clinical risk assessment, lifestyle analysis and population health research. The resulting dashboard provides a consistent way to compare disease indicators, clinical measurements, lifestyle categories, age groups, BMI categories and symptoms.

The project fulfils its descriptive visualisation objective by making the supplied data easier to explore and communicate. Its findings must be interpreted within the limits of the dataset: the dashboard is not a diagnostic or predictive tool, and observed associations should not be treated as causal evidence.

10. FUTURE SCOPE

- Add validated, representative datasets with clear data dictionaries and provenance.
- Include time-based records to study changes and trends over time.
- Investigate additional demographic and clinical variables where ethically and legally appropriate.
- Improve interactivity with carefully selected filters and drill-down views.
- Conduct formal usability testing with intended users and record feedback.
- If a predictive model is later introduced, validate it on suitable independent data and report appropriate evaluation metrics, fairness checks and limitations.
- Improve data quality documentation and refresh procedures.

11. APPENDIX

A. Source Code / Project Files

This is a Tableau-based analytics project rather than a conventional software application with a separate source-code repository. The principal project file is the Tableau workbook. Add the final saved workbook file name and repository location below.

Tableau workbook: health360.twbx

Additional files: Dataset CSV, report PDF, completed templates and demonstration video, subject to submission rules.

B. Dataset Link

Dataset file: https://drive.google.com/file/d/1RDbCMrWJaLQUufYcGbI6m9LgTvuJQcv2/view?usp=sharing

Dataset details: 5,000 records and 16 fields. The dataset was obtained through the SkillWallet capstone workspace. Insert the accessible dataset URL or the approved shared location here: https://drive.google.com/file/d/1RDbCMrWJaLQUufYcGbI6m9LgTvuJQcv2/view?usp=sharing

C. Project Demo Link

Tableau Public dashboard:
https://public.tableau.com/views/health360_17935475288100/health360project?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Project explanation video:https://drive.google.com/file/d/1sLCOIdEiCdBS9JFgbsjOPBC6P6HBfozw/view?usp=sharing

D. Dataset fields

The 16 fields identified in the dataset are: age, gender, bmi, blood_pressure, cholesterol, glucose, exercise_level, smoking, alcohol, fatigue, chest_pain, dizziness, heart_disease, diabetes, stroke and health_risk_score.

— End of Report —